

Table 12: Test RMSE of auNN and DGP for different values of M (number of inducing points) and D (number of hidden units) as the depth increases from $L = 2$ to $L = 4$. Mean and one standard error over 5 independent runs on the UCI Power dataset are shown. The lower the better.

L	D	M	auNN-RBF						auNN-TRI						DGP					
			5	10	25	50	75	100	5	10	25	50	75	100	5	10	25	50	75	100
2	5		4.20±0.09	4.14±0.08	4.12±0.08	4.13±0.08	4.13±0.07	4.14±0.09	4.16±0.08	4.09±0.08	4.06±0.09	4.12±0.09	4.32±0.13	4.12±0.09	4.24±0.10	4.19±0.09	4.08±0.09	4.05±0.08	4.03±0.07	4.01±0.07
	10		4.15±0.09	4.08±0.08	4.03±0.08	4.03±0.07	4.03±0.09	4.03±0.09	4.10±0.10	4.03±0.08	3.99±0.08	4.01±0.08	4.03±0.09	4.03±0.07	4.24±0.10	4.21±0.09	4.10±0.08	4.08±0.08	4.03±0.08	4.02±0.07
	25		4.09±0.08	4.01±0.08	4.01±0.08	4.01±0.08	4.00±0.09	4.02±0.08	4.04±0.08	3.96±0.07	3.90±0.08	3.91±0.08	3.89±0.08	3.92±0.07	4.24±0.09	4.18±0.09	4.10±0.09	4.06±0.08	4.03±0.08	4.01±0.08
	50		4.06±0.08	4.00±0.07	4.00±0.07	3.98±0.07	4.01±0.09	3.99±0.08	4.04±0.08	3.93±0.07	3.86±0.09	3.83±0.08	3.81±0.08	3.81±0.10	4.24±0.10	4.24±0.10	4.18±0.09	4.11±0.09	4.06±0.09	4.03±0.08
3	5		4.14±0.09	4.08±0.08	4.11±0.06	4.09±0.11	4.11±0.09	4.09±0.09	4.10±0.09	4.10±0.07	4.02±0.08	4.04±0.08	4.15±0.08	4.04±0.09	4.22±0.09	4.10±0.08	4.07±0.09	3.92±0.07	3.90±0.06	3.86±0.05
	10		4.02±0.08	4.02±0.08	4.00±0.07	4.02±0.07	4.02±0.07	3.99±0.07	4.01±0.08	3.95±0.07	3.92±0.07	3.92±0.08	3.90±0.07	3.90±0.08	4.20±0.09	4.10±0.08	3.98±0.08	3.99±0.07	4.05±0.08	4.03±0.08
	25		3.96±0.08	3.93±0.07	3.87±0.07	3.87±0.07	3.87±0.07	3.83±0.07	3.88±0.08	3.84±0.08	3.76±0.07	3.67±0.06	3.75±0.10	3.71±0.09	4.23±0.10	4.19±0.08	4.11±0.09	4.06±0.08	4.03±0.08	4.02±0.08
	50		3.89±0.08	3.88±0.07	3.85±0.06	3.80±0.09	3.82±0.07	3.80±0.08	3.86±0.08	3.77±0.09	3.62±0.06	3.61±0.08	3.59±0.09	3.60±0.07	4.24±0.10	4.24±0.10	4.12±0.09	4.07±0.08	4.04±0.08	4.03±0.08
4	5		4.14±0.10	4.10±0.08	4.08±0.08	4.11±0.09	4.07±0.05	4.09±0.09	12.00±3.26	4.04±0.07	4.06±0.09	4.07±0.07	4.09±0.08	4.07±0.08	4.22±0.09	4.10±0.08	3.97±0.07	3.93±0.07	3.88±0.07	3.85±0.07
	10		4.01±0.08	3.98±0.07	3.99±0.07	3.99±0.07	4.05±0.07	4.01±0.06	4.03±0.09	3.99±0.06	3.94±0.06	3.94±0.07	3.92±0.07	3.93±0.08	4.20±0.09	4.12±0.08	4.08±0.10	3.94±0.07	4.06±0.08	4.01±0.08
	25		3.93±0.09	3.91±0.08	3.87±0.07	3.78±0.06	3.84±0.07	3.82±0.07	3.94±0.07	3.85±0.08	3.76±0.08	3.75±0.08	3.77±0.10	3.78±0.09	4.24±0.09	4.18±0.09	4.11±0.09	4.06±0.08	4.04±0.08	4.03±0.08
	50		3.92±0.08	3.96±0.07	3.78±0.11	3.82±0.07	3.74±0.10	3.70±0.07	3.90±0.08	3.82±0.06	3.71±0.09	3.73±0.09	3.75±0.08	3.77±0.08	4.24±0.09	4.20±0.10	4.12±0.09	4.06±0.08	4.04±0.08	4.03±0.08

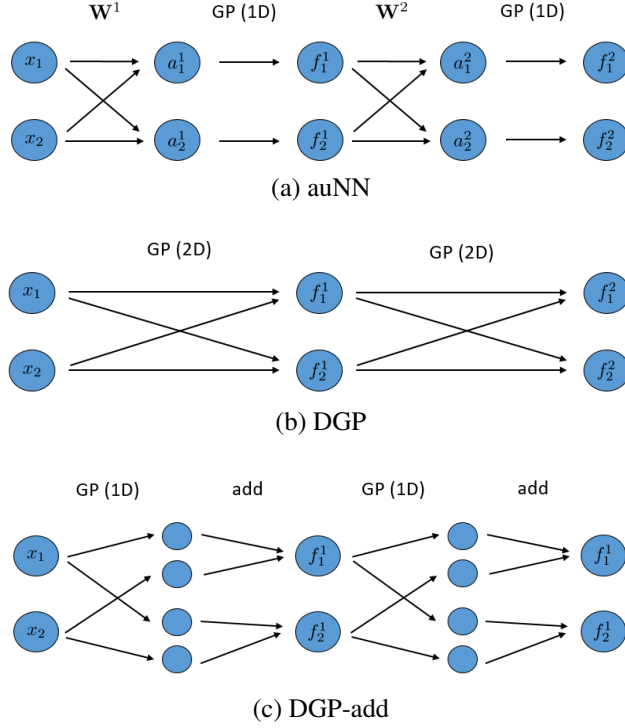


Figure 11: Representation of two hidden layers (with two units per layer) for auNN (a), DGP (b), and DGP-add (c).

D COMPUTATIONAL COST SUMMARY

Table 15 shows the training computational complexity for the methods compared in this paper. Moreover, in order to evaluate the computational cost in practice, the table also shows the actual running time for the experiment of Section 3.1. BNN is the fastest algorithm, since it utilizes a factorized Gaussian for the approximate posterior. Although fBNN has the same theoretical complexity, the Spectral Stein Gradient Estimator (Shi et al., 2018) is used to compute the KL divergence gradients. Moreover, a GP prior is specified at function space, for which a GP must be trained as a previous step. DGP and auNN have the same theoretical complexity. In practice, auNN is typically faster because it requires fewer inducing points, recall Section 3.3 and Table 3. The running time in Table 15 is very similar for both because the same amount of inducing points ($M = 10$) is used in this simple experiment.

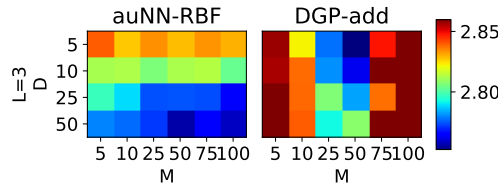


Figure 12: Test NLL on Power dataset for different values of D and M (the lower the better).

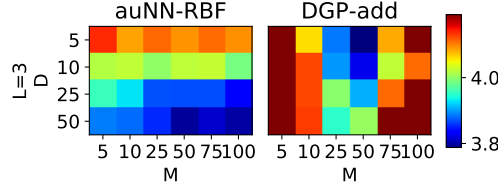
Figure 13: Test RMSE on Power dataset for different values of D and M (the lower the better).

Table 13: Standard error for the results in Table 2. Three random train-test splits are considered.

				auNN						DGP		
		N	D	RBF-2	RBF-3	RBF-4	TRI-2	TRI-3	TRI-4	DGP-2	DGP-3	DGP-4
Brier	HIGGS	11M	28	0.0001	0.0007	0.0008	0.0003	0.0005	0.0009	0.0018	0.0016	0.0006
	SUSY	5M	18	0.0005	0.0005	0.0006	0.0005	0.0005	0.0005	0.0011	0.0014	0.0021
ECE	HIGGS	11M	28	0.0015	0.0020	0.0022	0.0010	0.0035	0.0019	0.0006	0.0004	0.0008
	SUSY	5M	18	0.0012	0.0011	0.0014	0.0018	0.0012	0.0014	0.0005	0.0006	0.0008

Table 14: Standard error for the results in Table 3. Fifty independent runs are considered.

		auNN						DGP		
		RBF-2	RBF-3	RBF-4	TRI-2	TRI-3	TRI-4	DGP-2	DGP-3	DGP-4
HIGGS		0.0258	0.0325	0.0371	0.0188	0.0318	0.0378	0.0248	0.0266	0.0269
SUSY		0.0215	0.0274	0.0369	0.0202	0.0258	0.0350	0.0108	0.0126	0.0144

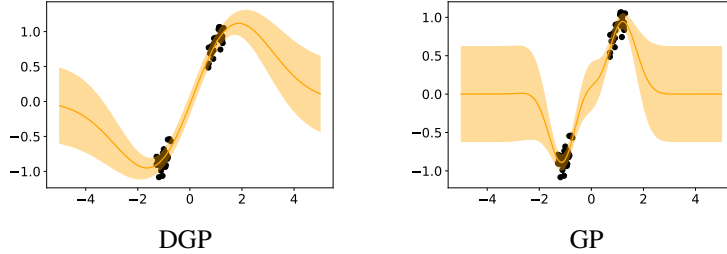


Figure 14: DGP and GP trained on the dataset of Section 3.1. The experimental details are analogous to those in Section 3.1, see Appendix B. Whereas DGP underestimates the uncertainty for in-between data, a simpler GP does provide increased uncertainty in the gap.

Table 15: Training computational cost for the models compared in this paper. The running time (in seconds) corresponds to the mean and one standard error over 10 independent runs of the experiment in Section 3.1. More details in Appendix D.

	BNN	fBNN	DGP	auNN
Running time (s)	15.21 ± 0.78	51.92 ± 1.07	22.37 ± 0.97	21.16 ± 0.89
Complexity	$\mathcal{O}(N \sum_i D^i D^{i+1})$	$\mathcal{O}(N \sum_i D^i D^{i+1})$	$\mathcal{O}(NM^2 \sum_i D^i)$	$\mathcal{O}(NM^2 \sum_i D^i)$